

Unit 3, Lesson H1: Plotting & Analyzing Residuals



Benchmark

MA.912.DP.2.5 Given a scatter plot that represents bivariate numerical data, assess the fit of a given linear function by plotting and analyzing residuals.



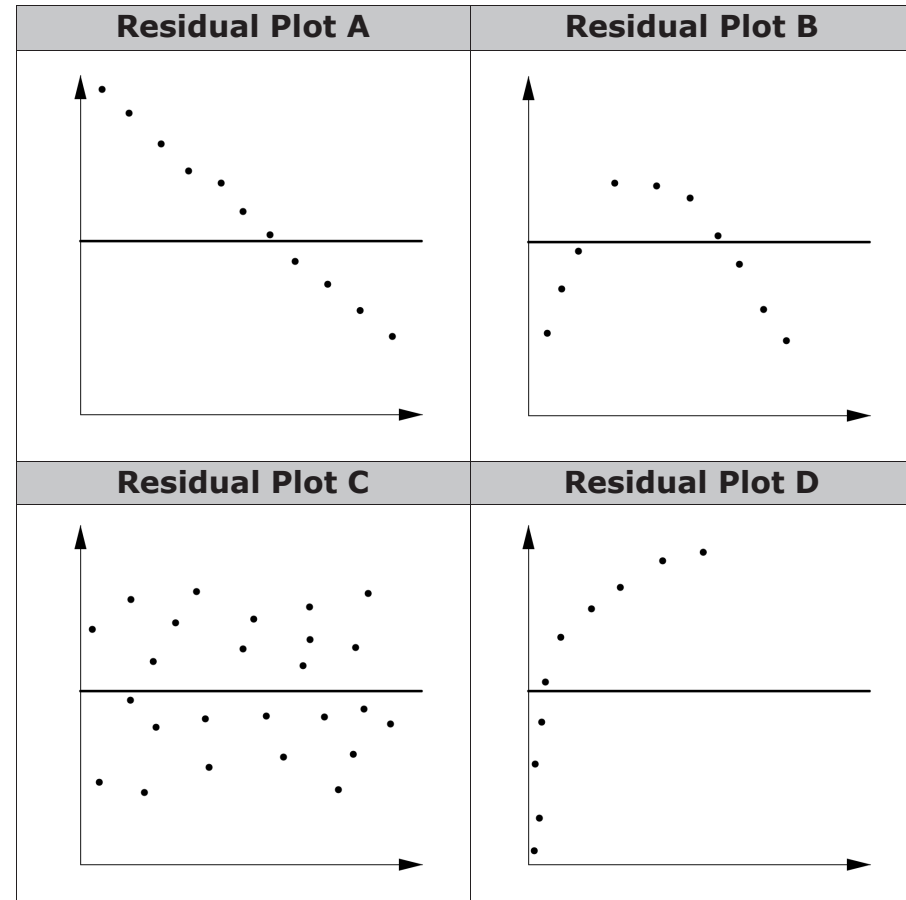
Learning Targets

- ☐ I can create a residual plot to assess the fit of a given linear function for bivariate data.
- ☐ I can determine whether a function is an appropriate fit for the data by analyzing a residual plot.



3.H1.1 Warm-Up: Which One Doesn't Belong?

Four residual plots are shown.



- Which residual plot does not belong?
- Explain your choice.

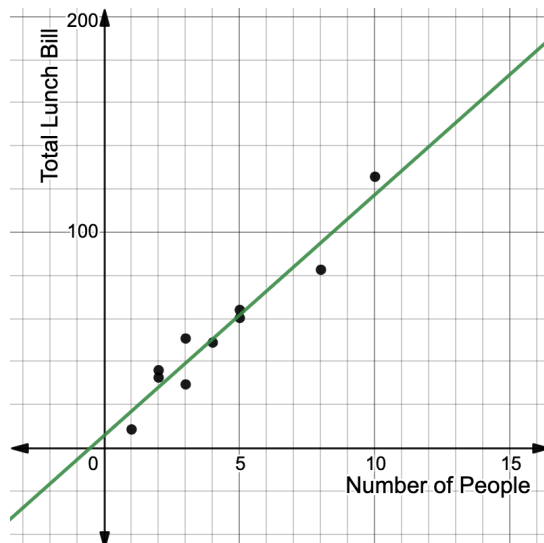


3.H1.2 Collaboration: Creating a Residual Plot

The owner of a local restaurant selected a random sample of lunch tables at her restaurant. For each table, the owner recorded the total amount of the lunch bill, before tip, and the number of people at the table.

Number of People	Total Lunch Bill
1	\$8.49
2	\$35.92
2	\$32.54
3	\$29.37
3	\$50.65
4	\$48.78
5	\$63.92
5	\$60.21
8	\$82.56
10	\$125.71

The owner's data is represented on the scatter plot with the function $B(n) = 5.82 + 11.16n$ fit to the data, where n is the number of people at a dinner party and B is the total amount of the lunch bill.



1. Work with your partner to use the linear function that models the relationship between the data and the actual data values to complete the table.

Number of People	Actual Total Bill	Predicted Total Bill	Residual Values
1	\$8.49	\$16.98	-8.49
2	\$35.92	\$28.14	
2	\$32.54	\$28.14	4.4
3	\$29.37		-9.93
3	\$50.65		
4	\$48.78	\$50.46	-1.68
5	\$63.92	\$61.62	2.30
5	\$60.21	\$61.62	-0.41
8	\$82.56	\$95.10	-12.54
10	\$125.71		

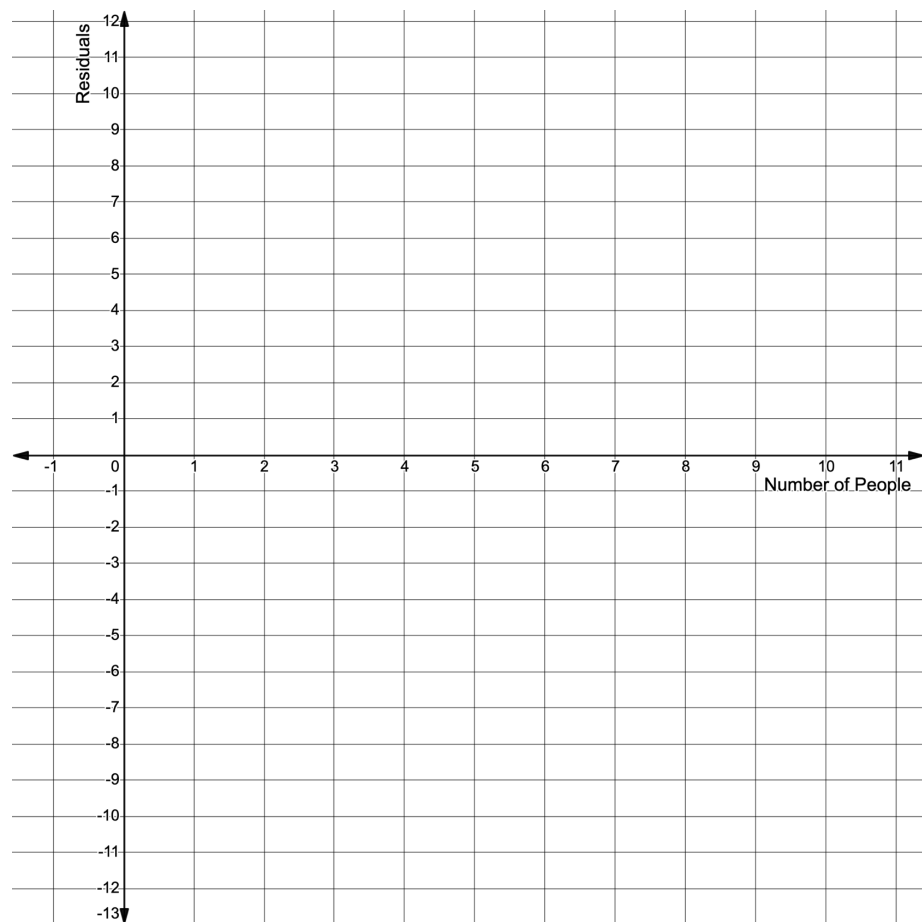
A **residual** is the difference between an actual data value and the predicted value. It is the vertical distance from the data point to the line of best fit.

$$\text{Residual} = (\text{actual } y\text{-value}) - (\text{predicted } y\text{-value})$$

2. Use your calculations to complete the table.

Number of Positive Residuals	Number of Negative Residuals	Residual with the Smallest Absolute Value	Residual with the Largest Absolute Value

3. Work with your partner to plot the residual values on the coordinate grid shown.

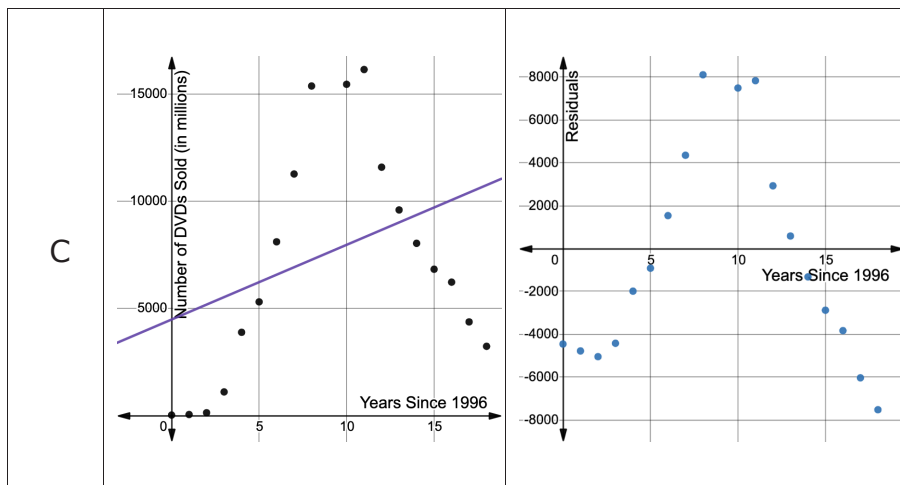


3.H1.3 Exploration: Determining the Fit of a Function by Analyzing Residuals

Work with your partner to complete the following.

Three scatter plots are shown, each fit with a linear function. The residual plots comparing the actual data values on the scatter plot to the predicted values using the linear functions are also shown for each set of bivariate data.

Data Set	Scatter Plot	Residual Plot
A		
B		

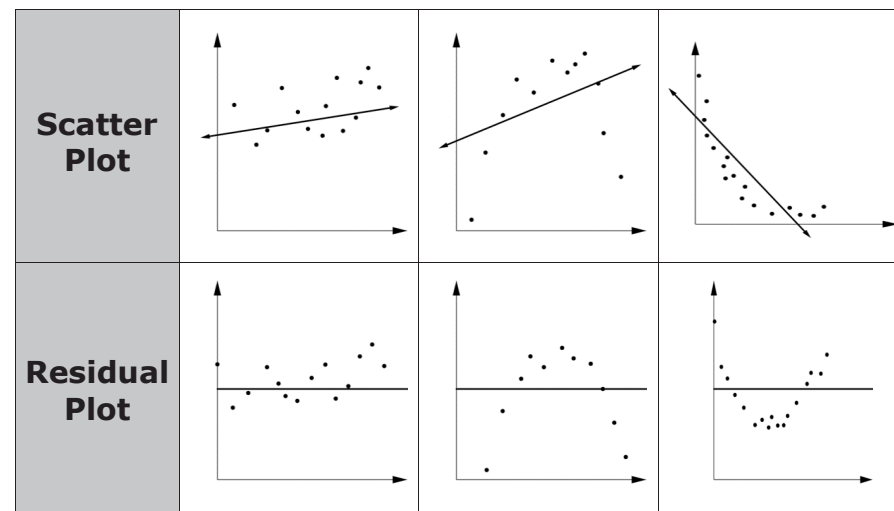


1. Work with your partner to explain which scatter plot appears to show the strongest linear relationship.
2. Describe how the residual plot for the linear function fit to the scatter plot selected in the previous question appears.
3. Which phrase best describes the residual plots for the functions that model relationships that do not appear to be linear?
 - The residual plot includes points above and below the x -axis that appear random and not following a pattern.
 - The residual plot includes points that appear to follow a pattern.



3.H1.4 Bring It Together: Determining the Fit of a Function by Analyzing Residuals

Three scatter plots with a linear function fit to the data and their corresponding residual plots are shown.



The residual plot can be used to determine whether or not the function is a good fit.

1. Complete the statements.

If the function is a good fit for modeling the relationship between variables, the residual plot should include

- points that follow a pattern
 - points that are randomly scattered
- and

- a balance of points above and below the x -axis.
- most of the points on one side of the x -axis.

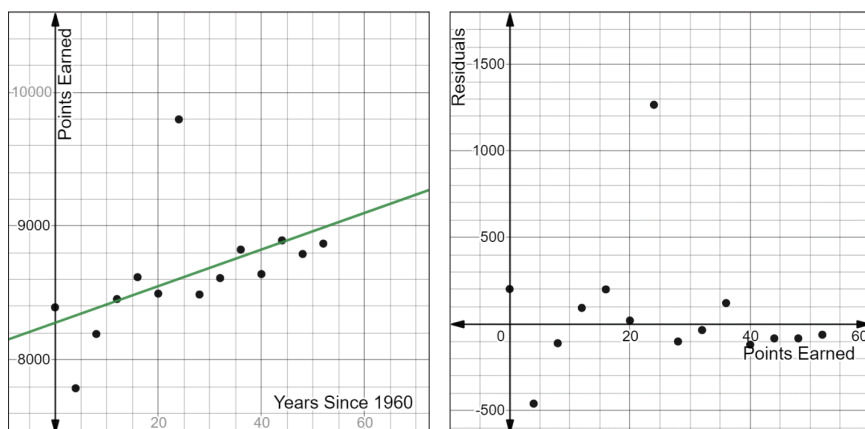
The better the function fits the data, the ○ closer
○ farther away
 the residual values will be ○ to
○ from the x -axis.

Residual plots can also be used to identify outliers in the data set.



An **outlier** is a value that is much higher or much lower than the other values in a set of data.

- The scatter plot with a linear function fit to the data and its corresponding residual plot are shown. The scatter plot represents the relationship between the number of points earned by men's decathlon Olympic gold medalists and years since 1960.



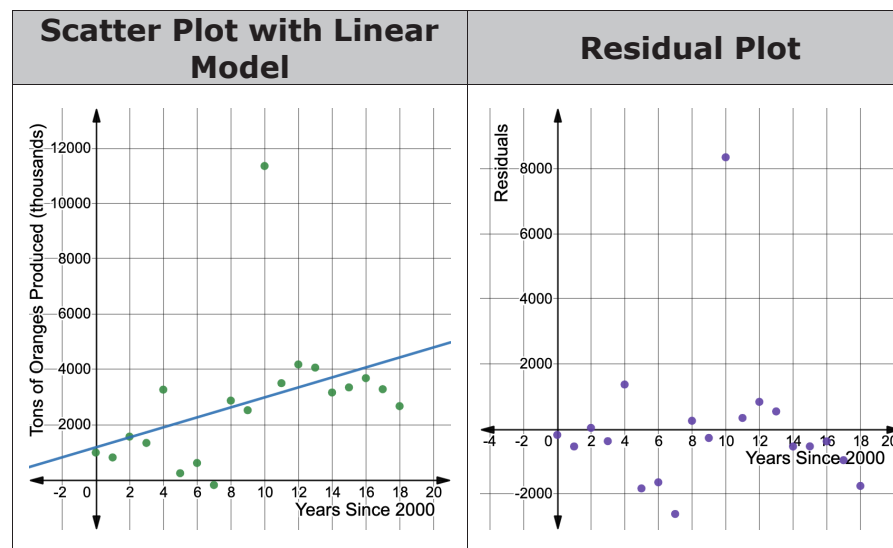
- Circle the point on the scatter plot that appears to be an outlier and its corresponding residual value on the residual plot.
- Complete the following statement.

In a set of bivariate numerical data, the residual of an outlier has the ☐ smallest ☐ greatest absolute value when compared to other residual values.



3.H1.5 Wrap-Up: Ticket Out the Door

One of the many things that Florida is known for are its oranges. Oranges, however, are grown and harvested in multiple states across the U.S. Below is a scatter plot representing U.S. orange production since 2000. A linear function is fit to the data, and the corresponding residual plot is shown.



- Explain whether or not the linear function is a good fit for this data set.
- Explain whether or not there appears to be an outlier in the data set using the residual plot.